

DEUTSCHE BANK INDIA

Retail Banking — Digital Payments

USE CASE CATALOGUE

UPI Enablement Platform

Actor-Goal Use Cases Derived from BRD v1.0 and FRS v1.0

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Prepared by: Aniket Saraf

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1. Introduction

This Use Case catalogue elaborates the functional requirements defined in the Business Requirements Document (BRD v1.0) and Functional Requirements Specification (FRS v1.0) for the Deutsche Bank India UPI Enablement Platform into actor-goal-oriented use cases. Each use case identifies the primary actor, preconditions, trigger, main success scenario, alternate/exception flows, postconditions, and traceability to BRD/FRS requirement IDs.

1.1 Actors

Actor	Description
Retail Customer	A Deutsche Bank India retail banking customer using the mobile app
Payer	The retail customer acting as the paying party in a transaction or collect request
Requester / Payee	The retail customer requesting or receiving payment
Operations User (Tier 2)	Internal Operations staff investigating transactions and disputes
System (Scheduler)	Automated batch/scheduled jobs (expiry, reconciliation, escalation)
NPCI / PSP Switch	External system participant in transaction routing (not a human actor, included for flow completeness)

1.2 Use Case Catalogue Index

ID	Use Case	Module
UC-01	Register for UPI	Customer Registration
UC-02	Choose Custom UPI ID	Customer Registration
UC-03	Link Additional Account	Bank Account Linking
UC-04	Remove Linked Account	Bank Account Linking
UC-05	Change Default Account	Bank Account Linking
UC-06	Set UPI PIN	UPI PIN Management
UC-07	Reset UPI PIN	UPI PIN Management
UC-08	Change UPI PIN	UPI PIN Management
UC-09	Handle PIN Lockout	UPI PIN Management
UC-10	Send Money via Mobile Number	Send Money
UC-11	Send Money via UPI ID	Send Money
UC-12	Send Money via Bank Account + IFSC	Send Money
UC-13	Self Transfer Between Own Accounts	Send Money
UC-14	Reconcile Failed/Pending Transaction	Send Money
UC-15	Create Collect Request	Collect Requests
UC-16	Approve Collect Request	Collect Requests
UC-17	Reject Collect Request	Collect Requests

ID	Use Case	Module
UC-18	Auto-Expire Collect Request	Collect Requests
UC-19	Generate QR Code	QR Payments
UC-20	Scan & Pay via QR	QR Payments
UC-21	Check Account Balance	Balance Check
UC-22	View and Filter Transaction History	Transaction History
UC-23	Export Transaction Statement to PDF	Transaction History
UC-24	Receive Real-Time Notification	Notifications
UC-25	Manage Notification Preferences	Notifications
UC-26	Raise Transaction Dispute	Dispute Management
UC-27	Track Dispute Status	Dispute Management
UC-28	Investigate and Update Dispute	Dispute Management
UC-29	Auto-Escalate Overdue Dispute	Dispute Management

2. Use Cases — Modules 1 to 5

UC-01 — Register for UPI

Module: Customer Registration

Field	Detail
Actor	Retail Customer
Preconditions	Customer has an active mobile banking login and at least one KYC-compliant, active savings/current account
Trigger	Customer selects 'Register for UPI' from the app menu
Main Success Scenario	1. Customer initiates registration. 2. System verifies mobile number via OTP. 3. System captures data-sharing consent. 4. System fetches eligible accounts from CBS/CIF. 5. Customer selects account(s) and designates a default. 6. System creates the UPI profile and assigns a VPA. 7. System displays success confirmation and prompts PIN setup.
Alternate Flows	A1 (at step 5): Customer selects only one account — it is auto-designated default. A2: Customer chooses to customise the VPA handle (see UC-02).
Exception Flows	E1 (at step 2): Mobile number does not match CBS/CIF — registration halted with remediation message. E2 (at step 3): Consent declined — registration halted. E3 (at step 4): No eligible accounts found — customer directed to branch/support.
Postconditions	UPI profile created and linked to Customer ID; audit log entry created; registration-confirmation notification sent
Related Requirements	FR-REG-01 to FR-REG-12; BR-REG-01 to BR-REG-07

UC-02 — Choose Custom UPI ID

Module: Customer Registration

Field	Detail
Actor	Retail Customer
Preconditions	Customer is mid-registration, past mobile verification and consent
Trigger	Customer selects 'Customise UPI ID' during registration
Main Success Scenario	1. Customer enters desired VPA handle. 2. System checks availability via PSP switch/NPCI namespace. 3. Handle is available and reserved. 4. Registration proceeds with the custom VPA.
Alternate Flows	A1: System suggests alternate available handles if the requested one is taken.
Exception Flows	E1: Handle contains reserved/profane words — rejected with guidance.
Postconditions	Custom VPA reserved and assigned to the UPI profile
Related Requirements	FR-REG-10

UC-03 — Link Additional Account

Module: Bank Account Linking

Field	Detail
Actor	Retail Customer
Preconditions	Customer has an active UPI profile and at least one additional eligible account not yet linked
Trigger	Customer selects 'Add Account' from Manage Linked Accounts
Main Success Scenario	1. System fetches remaining eligible accounts. 2. Customer selects an account to link. 3. System creates a Linked_Account record. 4. Confirmation displayed.
Exception Flows	E1: Selected account is an ineligible type — blocked (ERR-ACC-002). E2: Account already linked — blocked (ERR-ACC-003).
Postconditions	New Linked_Account record created; audit log entry created
Related Requirements	FR-ACC-01, FR-ACC-03, FR-ACC-08; BR-ACC-01

UC-04 — Remove Linked Account

Module: Bank Account Linking

Field	Detail
Actor	Retail Customer
Preconditions	Customer has more than one linked account, or is removing a non-default account
Trigger	Customer selects 'Remove' on a linked account
Main Success Scenario	1. Customer selects account to remove. 2. System confirms it is not the default. 3. System sets link_status = Delinked. 4. Confirmation displayed.
Exception Flows	E1: Customer attempts to remove the default account while other accounts remain linked — blocked with ERR-ACC-001, prompting the customer to set a new default first.
Postconditions	Linked_Account.link_status = Delinked; audit log entry created
Related Requirements	FR-ACC-04; BR-ACC-04

UC-05 — Change Default Account

Module: Bank Account Linking

Field	Detail
Actor	Retail Customer
Preconditions	Customer has two or more linked accounts
Trigger	Customer selects 'Set as Default' on a non-default linked account
Main Success Scenario	1. Customer selects a non-default account. 2. System sets is_default = true on the new selection and false on the prior default within a single atomic transaction. 3. Confirmation displayed.
Postconditions	Default account updated across Linked_Account records
Related Requirements	FR-ACC-02, FR-ACC-05

3. Use Cases — UPI PIN Management

UC-06 — Set UPI PIN

Module: UPI PIN Management

Field	Detail
Actor	Retail Customer
Preconditions	Customer has an active UPI profile without a PIN set
Trigger	Customer selects 'Set PIN' after registration or from settings
Main Success Scenario	1. Customer validates identity via debit card (last 6 digits + expiry) or Aadhaar-OTP. 2. System sends and validates mobile OTP. 3. Customer enters and confirms new PIN via secure pad. 4. System encrypts PIN on-device and submits via PSP/NPCI PIN-set flow. 5. Confirmation displayed.
Exception Flows	E1: Card details mismatch (ERR-PIN-001). E2: Card expired (ERR-PIN-002). E3: OTP invalid/expired (ERR-PIN-003). E4: New PIN and Confirm PIN mismatch — re-entry prompted.
Postconditions	UPI_PIN record created; audit log entry created (no PIN value logged); confirmation notification sent
Related Requirements	FR-PIN-01 to FR-PIN-04; BR-PIN-01, BR-PIN-02

UC-07 — Reset UPI PIN

Module: UPI PIN Management

Field	Detail
Actor	Retail Customer
Preconditions	Customer has forgotten their PIN or is unlocking a locked profile
Trigger	Customer selects 'Forgot PIN' / 'Reset PIN'
Main Success Scenario	1. Customer validates identity via debit card/OTP (as in UC-06, without requiring old PIN). 2. Customer sets a new PIN. 3. System resets failed_attempt_count to 0 and clears lock_status if applicable.
Postconditions	PIN reset; profile unlocked if previously locked
Related Requirements	FR-PIN-05; BR-PIN-04

UC-08 — Change UPI PIN

Module: UPI PIN Management

Field	Detail
Actor	Retail Customer
Preconditions	Customer has an active, unlocked PIN
Trigger	Customer selects 'Change PIN' from settings
Main Success Scenario	1. Customer enters existing PIN. 2. System validates via PSP/NPCI PIN-verify call. 3. Customer sets and confirms new PIN. 4. Confirmation displayed.
Exception Flows	E1: Existing PIN incorrect — standard incorrect-PIN handling applies (see UC-09).

Field	Detail
Postconditions	PIN updated
Related Requirements	FR-PIN-06; BR-PIN-05

UC-09 — Handle PIN Lockout

Module: UPI PIN Management

Field	Detail
Actor	Retail Customer; System
Preconditions	Customer has entered an incorrect PIN during any PIN-authenticated action
Trigger	Third consecutive incorrect PIN entry
Main Success Scenario	1. System increments failed_attempt_count on each incorrect entry. 2. On the 3rd consecutive failure, system sets lock_status = Locked and blocks all PIN-authenticated actions. 3. System sends a security alert notification. 4. Profile auto-unlocks after 24 hours, or customer resets PIN (UC-07) for immediate unlock.
Postconditions	Profile locked/unlocked per outcome; security notification sent
Related Requirements	FR-PIN-07, FR-PIN-08; BR-PIN-03

4. Use Cases — Send Money

UC-10 — Send Money via Mobile Number

Module: Send Money

Field	Detail
Actor	Retail Customer (Payer)
Preconditions	Customer is UPI-registered with a set PIN and at least one linked account
Trigger	Customer selects Send Money → Mobile Number
Main Success Scenario	1. Customer enters recipient mobile number. 2. System resolves the number to a VPA via NPCI. 3. Customer enters amount and optional note. 4. Customer authorises with UPI PIN. 5. System performs balance, fraud, and limit checks. 6. System submits to NPCI via PSP switch. 7. System displays real-time status and sends notification.
Alternate Flows	A1 (step 2): Multiple VPAs found for the number — customer selects the intended one.
Exception Flows	E1: Number not UPI-registered (ERR-TXN-001). E2: Insufficient balance (ERR-TXN-007). E3: Limit breach (ERR-TXN-005/006/009). E4: Fraud decline (ERR-TXN-008). E5: NPCI timeout — transaction marked Pending, reconciliation triggered (see UC-14).
Postconditions	Transaction recorded with terminal status; account balance updated; notification sent
Related Requirements	FR-TXN-01, FR-TXN-06 to FR-TXN-11; BR-TXN-01 to BR-TXN-04

UC-11 — Send Money via UPI ID

Module: Send Money

Field	Detail
Actor	Retail Customer (Payer)
Preconditions	Customer is UPI-registered with a set PIN
Trigger	Customer selects Send Money → UPI ID
Main Success Scenario	1. Customer enters recipient VPA. 2. System validates format and resolves via PSP switch, displaying the recipient name. 3. Customer confirms, enters amount, authorises with PIN. 4. System performs standard checks and submits to NPCI. 5. Status displayed and notification sent.
Exception Flows	E1: VPA does not resolve (ERR-TXN-002). E2: Recipient blacklisted (ERR-TXN-004). Standard balance/limit/fraud exceptions per UC-10 also apply.
Postconditions	Transaction recorded with terminal status
Related Requirements	FR-TXN-02, FR-TXN-06 to FR-TXN-11

UC-12 — Send Money via Bank Account + IFSC

Module: Send Money

Field	Detail
Actor	Retail Customer (Payer)

Field	Detail
Preconditions	Customer is UPI-registered with a set PIN
Trigger	Customer selects Send Money → Bank Account + IFSC
Main Success Scenario	1. Customer enters account number and IFSC. 2. System validates IFSC against the IFSC master. 3. System initiates an account-based UPI transfer via PSP switch. 4. Customer authorises with PIN. 5. Standard checks apply; status displayed.
Exception Flows	E1: Invalid IFSC (ERR-TXN-003). Standard balance/limit/fraud exceptions per UC-10 also apply.
Postconditions	Transaction recorded with terminal status
Related Requirements	FR-TXN-04, FR-TXN-06 to FR-TXN-11

UC-13 — Self Transfer Between Own Accounts

Module: Send Money

Field	Detail
Actor	Retail Customer
Preconditions	Customer has two or more linked accounts
Trigger	Customer selects Send Money → Self Transfer
Main Success Scenario	1. Customer selects source and destination from own linked accounts. 2. Customer enters amount and authorises with PIN. 3. System applies per-transaction/daily limits (payee-specific limits do not apply) and submits transaction. 4. Status displayed.
Postconditions	Transaction recorded; both account balances updated
Related Requirements	FR-TXN-05; BR-TXN-07

UC-14 — Reconcile Failed/Pending Transaction

Module: Send Money

Field	Detail
Actor	System (Scheduler)
Preconditions	A transaction has timed out without a definitive NPCI response
Trigger	Transaction submission exceeds the NPCI SLA response window
Main Success Scenario	1. System marks the transaction Pending. 2. Deferred Status Check job polls NPCI/CBS reconciliation status at defined intervals. 3. On confirmation of successful credit, transaction is marked Success. 4. On confirmation of debit without credit, system triggers automatic reversal and notifies the customer.
Postconditions	Transaction reaches a terminal status (Success/Failed/Reversed); customer notified
Related Requirements	FR-TXN-14; BR-TXN-08; NFR-AVL-04

5. Use Cases — Collect Requests

UC-15 — Create Collect Request

Module: Collect Requests

Field	Detail
Actor	Retail Customer (Requester)
Preconditions	Customer is UPI-registered
Trigger	Customer selects 'Request Money'
Main Success Scenario	1. Customer enters payer VPA/mobile, amount, optional note. 2. System resolves payer and creates a Collect_Request with status Pending and 24-hour expiry. 3. System notifies the payer in real time.
Postconditions	Collect_Request created; payer notified
Related Requirements	FR-COL-01, FR-COL-02; BR-COL-01, BR-COL-03

UC-16 — Approve Collect Request*Module: Collect Requests*

Field	Detail
Actor	Retail Customer (Payer)
Preconditions	A Pending collect request exists addressed to the customer
Trigger	Customer opens and approves the request
Main Success Scenario	1. System displays requester, amount, note. 2. Customer authorises with UPI PIN. 3. System processes the debit as a standard payment (per UC-10/11 checks). 4. Collect_Request.status set to Approved.
Exception Flows	E1: Request already expired (ERR-COL-001). E2: Request already actioned (ERR-COL-002). Standard balance/limit/fraud exceptions also apply.
Postconditions	Payment processed; Collect_Request marked Approved
Related Requirements	FR-COL-03; BR-COL-02

UC-17 — Reject Collect Request*Module: Collect Requests*

Field	Detail
Actor	Retail Customer (Payer)
Preconditions	A Pending collect request exists addressed to the customer
Trigger	Customer selects Reject
Main Success Scenario	1. Customer optionally enters a reason. 2. System sets status to Rejected. 3. Requester is notified with the outcome and reason if provided.
Postconditions	Collect_Request marked Rejected; requester notified
Related Requirements	FR-COL-04; BR-COL-05

UC-18 — Auto-Expire Collect Request*Module: Collect Requests*

Field	Detail
Actor	System (Scheduler)
Preconditions	A Pending collect request has reached its 24-hour expiry
Trigger	Scheduled expiry job runs (minimum hourly)
Main Success Scenario	1. Job identifies Pending requests past expiry_at. 2. Status set to Expired. 3. Requester notified.
Postconditions	Collect_Request marked Expired
Related Requirements	FR-COL-05; BR-COL-01

6. Use Cases — QR Payments

UC-19 — Generate QR Code

Module: QR Payments

Field	Detail
Actor	Retail Customer
Preconditions	Customer is UPI-registered with at least one linked account
Trigger	Customer selects 'My QR Code'
Main Success Scenario	1. Customer selects Static or Dynamic type (and amount/validity for Dynamic). 2. System constructs the QR payload per NPCI specification and renders the image. 3. Customer may share/download the QR.
Postconditions	QR_Code record created; QR image rendered
Related Requirements	FR-QR-01, FR-QR-02, FR-QR-05; BR-QR-01, BR-QR-02

UC-20 — Scan & Pay via QR

Module: QR Payments

Field	Detail
Actor	Retail Customer (Payer)
Preconditions	Customer is UPI-registered with a set PIN
Trigger	Customer selects Scan & Pay
Main Success Scenario	1. Camera captures and decodes the QR payload. 2. For merchant QR, system validates against the NPCI merchant registry. 3. Customer confirms/enters amount (if not embedded) and authorises with PIN. 4. Standard transaction checks apply and status is displayed.
Exception Flows	E1: QR unreadable (ERR-QR-001). E2: Merchant unverified/blacklisted (ERR-QR-002). E3: Dynamic QR expired (ERR-QR-003).
Postconditions	Transaction recorded with terminal status
Related Requirements	FR-QR-03, FR-QR-04, FR-QR-06; BR-QR-03, BR-QR-04

7. Use Cases — Balance Check

UC-21 — Check Account Balance

Module: Balance Check

Field	Detail
Actor	Retail Customer
Preconditions	Customer is UPI-registered with a set PIN; daily enquiry cap not yet reached
Trigger	Customer selects 'Check Balance'
Main Success Scenario	1. Customer selects account. 2. Customer authorises with PIN. 3. System validates the daily cap, then submits enquiry via PSP switch. 4. Available and current balance displayed.
Exception Flows	E1: Daily cap exceeded (ERR-BAL-001).
Postconditions	Balance displayed; no persistent client-side cache retained
Related Requirements	FR-BAL-01 to FR-BAL-03; BR-BAL-01, BR-BAL-02

8. Use Cases — Transaction History

UC-22 — View and Filter Transaction History

Module: Transaction History

Field	Detail
Actor	Retail Customer
Preconditions	Customer is UPI-registered and has at least one transaction
Trigger	Customer opens Transaction History
Main Success Scenario	1. System displays the most recent 6 months of transactions by default. 2. Customer applies filters (date, status, amount, beneficiary, type). 3. System returns matching results. 4. Customer drills into a transaction for full detail.
Postconditions	None (read-only)
Related Requirements	FR-HIS-01, FR-HIS-02, FR-HIS-04, FR-HIS-05; BR-HIS-01

UC-23 — Export Transaction Statement to PDF

Module: Transaction History

Field	Detail
Actor	Retail Customer
Preconditions	Customer is viewing Transaction History
Trigger	Customer selects 'Export PDF'
Main Success Scenario	1. Customer selects a date range. 2. System generates a PDF statement matching active filters. 3. Customer downloads or shares the file.
Alternate Flows	A1: For ranges exceeding 12 months, generation is processed asynchronously with an in-app notification when ready.
Postconditions	PDF statement generated
Related Requirements	FR-HIS-03

9. Use Cases — Notifications

UC-24 — Receive Real-Time Notification

Module: Notifications

Field	Detail
Actor	Retail Customer; System
Preconditions	A notifiable event occurs (payment, collect, refund, registration/PIN, security)
Trigger	Event status determined (e.g., transaction completes)
Main Success Scenario	1. Event published to Notification Service. 2. Message composed from the versioned template. 3. Notification dispatched via Push and/or SMS within 5 seconds. 4. Delivery status logged.

Field	Detail
Alternate Flows	A1: Push delivery fails for a security-critical event — system falls back to SMS.
Postconditions	Notification record created with delivery status
Related Requirements	FR-NOT-01 to FR-NOT-06; BR-NOT-01

UC-25 — Manage Notification Preferences

Module: Notifications

Field	Detail
Actor	Retail Customer
Preconditions	Customer is UPI-registered
Trigger	Customer opens Notification Preferences
Main Success Scenario	1. Customer toggles marketing/cross-sell notifications on or off. 2. System saves the preference and applies it to future non-mandatory notifications.
Postconditions	Preference updated; transaction and security notifications remain unaffected
Related Requirements	FR-NOT-07; BR-NOT-02

10. Use Cases — Dispute Management

UC-26 — Raise Transaction Dispute

Module: Dispute Management

Field	Detail
Actor	Retail Customer
Preconditions	Customer has a completed transaction within the last 180 days
Trigger	Customer selects 'Raise Dispute' from Transaction Detail
Main Success Scenario	1. Customer selects a dispute category and enters a description, with optional evidence attachment. 2. System generates an internal reference number. 3. System routes the dispute for investigation and, where applicable, escalates to NPCI, capturing the NPCI reference number.
Exception Flows	E1: Transaction outside the 180-day window (ERR-DIS-001). E2: A dispute is already open for this transaction (ERR-DIS-002).
Postconditions	Dispute record created with Submitted status; internal reference number issued
Related Requirements	FR-DIS-01 to FR-DIS-04; BR-DIS-01, BR-DIS-02

UC-27 — Track Dispute Status

Module: Dispute Management

Field	Detail
Actor	Retail Customer
Preconditions	Customer has at least one raised dispute

Field	Detail
Trigger	Customer opens Dispute Tracking
Main Success Scenario	1. System displays current status (Submitted / Under Investigation / Escalated / Resolved). 2. Where applicable, refund status and expected/actual credit date are displayed.
Postconditions	None (read-only)
Related Requirements	FR-DIS-05, FR-DIS-06

UC-28 — Investigate and Update Dispute

Module: Dispute Management

Field	Detail
Actor	Operations User (Tier 2)
Preconditions	A dispute exists in Submitted or Under Investigation status
Trigger	Operations user opens the dispute queue
Main Success Scenario	1. Operations user reviews dispute detail and evidence. 2. User adds investigation notes (visible to Compliance/Audit). 3. User updates status, and, if upheld, initiates a Refund record.
Postconditions	Dispute_Note created; Dispute status updated; Refund record created if applicable; customer notified
Related Requirements	FR-DIS-08

UC-29 — Auto-Escalate Overdue Dispute

Module: Dispute Management

Field	Detail
Actor	System (Scheduler)
Preconditions	A dispute remains unresolved beyond its category-specific NPCI TAT threshold
Trigger	Scheduled escalation job runs
Main Success Scenario	1. Job identifies disputes exceeding TAT. 2. Dispute is flagged for supervisor escalation. 3. Where applicable, dispute is auto-submitted to NPCI escalation channels.
Postconditions	Dispute status set to Escalated; supervisor notified
Related Requirements	FR-DIS-07; BR-DIS-03